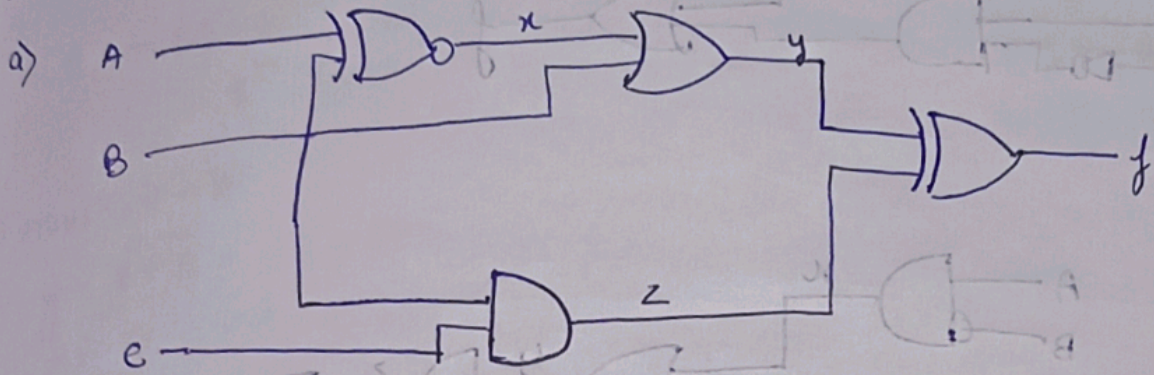


② Boolean Algebra - Assignment
i.e. output expression for

② Boolean
 a) Derive the output expression for the below circuits and draw the simplified circuits.



$$\Rightarrow x = \overline{A} \overline{B} + AB$$

$$y = \overline{A}\overline{B} + AB + B$$

22 BG

$$f^2 y \oplus z$$

$$f = (\overline{AB} + AB + B)(\overline{BC}) + (\overline{AB} + AB + B)(\overline{BC})$$

$$f^2 = (\overline{A}\overline{B})(\overline{A}B)(\overline{B})(B) + (\overline{A}B + B(A+1))(\overline{B} + \overline{C})$$

$$f = (\bar{A} + \bar{B})(\bar{A} + \bar{B})(\bar{B})(BC) + (\bar{A}\bar{B} + B)(\bar{B} + \bar{C})$$

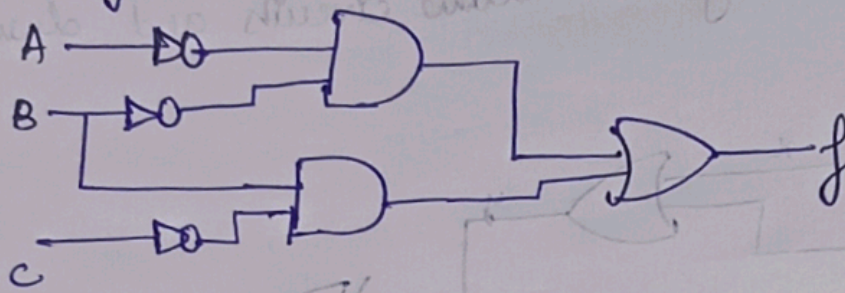
$$f = (A+B)(\bar{A}+\bar{B})0 + (\bar{A}\bar{B} + \bar{A}\bar{B}\bar{C} + 0 + \bar{B}\bar{C})$$

$$f = \overline{A}\overline{B}(1 + \overline{C}) + B\overline{C} \quad (9 + 0) = 8 \quad (8 + \overline{8A}) = 8$$

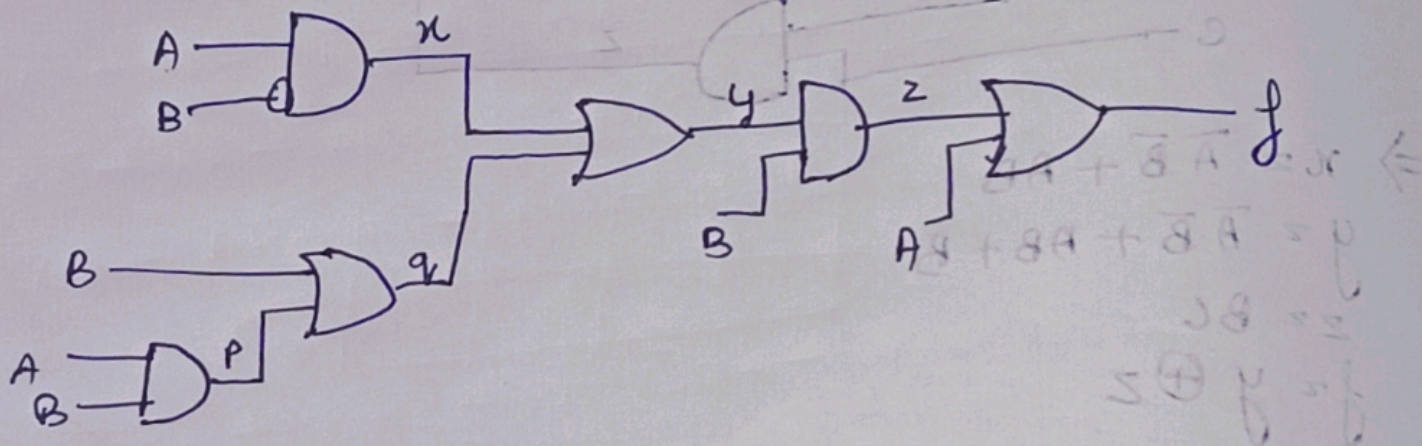
$$f = \overline{A} \overline{B} + B \overline{C}$$

$$\therefore f = \underline{\underline{\overline{A} \overline{B} + B \overline{C}}}$$

Simplified circuit



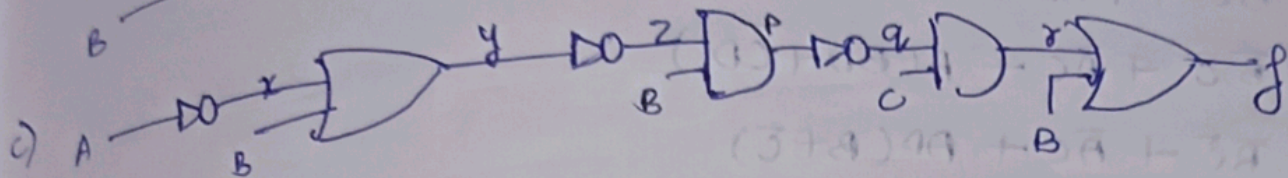
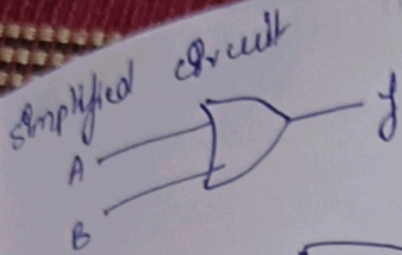
b)



now $x = AB$
 $P = AB$
 $q = B + AB$
 $y = x + q$
 $y = AB + B$
 $z = yB$
 $z = (AB + B)B = (0 + B) = B$

$f = z + A$

$\therefore f = B + A$



now $x = \bar{A}$

$$y = x + B = (\bar{A} + B)$$

$$z = \bar{y} = \overline{(\bar{A} + B)} = \bar{A} \bar{B} = \bar{A} \bar{B}$$

$$p = zB = (\bar{A} \bar{B})B = 0$$

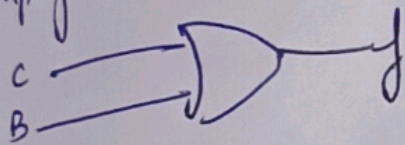
$$q = \bar{p} = 1$$

$$r = qC = C$$

$$f = r + B = C + B$$

$$\therefore f = C + B$$

simplified circuit



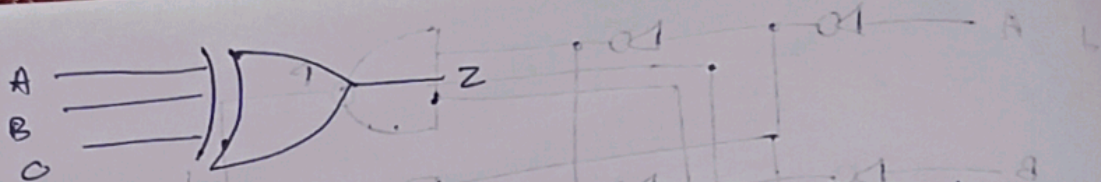
2) Minimize the following Boolean expression using Boolean postulates

$$a) \bar{A} \bar{C} \bar{D} E + \bar{A} \bar{B} \bar{D} + ABCE + ABD$$

$$= \bar{A} \bar{D} (CE + \bar{B}) + AB(CE + D)$$

cannot be further simplified

$$b) \bar{A} \bar{B} \bar{C} + ABD + \bar{A} C + \bar{A} C \bar{D} + \bar{A} C D + \bar{A} \bar{B} \bar{C}$$



4) minimize the following Boolean expression using K-Map

a) $y(a, b, c, d) = \sum m(4, 6, 8, 10, 11, 12, 15) + \sum d(3, 5, 7, 9)$

3

ab \ cd	00	01	11	10
00	0	0	X	0
01	1	X	X	1
11	1	0	1	0
10	1	X	1	1

$$y = cd + \bar{a}b + a\bar{b} + a\bar{c}\bar{d}$$

b) $y(a, b, c, d) = \sum m(0, 2, 4, 6, 7, 9) + \sum d(10, 11)$

ab \ cd	00	01	10	11
00	0	1	1	0
01	0	1	0	0
10	1	1	1	1
11	1	0	X	X

$$y = a\bar{d} + ab + \bar{a}\bar{b}d + \bar{a}\bar{c}d$$